

Production Order List — CSV Export — Basic Design Document (基本設計書)

Document control (改版履歴)

Field	Value
Document ID	002_BD-CSV
Category	UI + API (file download)
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-016
Based on brief.md revision	1
Created by	Claude (for ThongTM)
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Last updated date	2026-10-07

Version	Date	Author	Revision content
1	2026-10-07	Claude (for ThongTM)	Initial creation: 「CSV出力」 (Export CSV) on SCR-002 (WI-016 DEC-001–DEC-008)
2	2026-10-07	Claude (for ThongTM)	Review: the export keeps using the applied (searched) filters; a visible hint, item 30, now says so next to the button (WI-016 DEC-009). Wireframes updated

This document adds one action to SCR-002 製造指示一覧 (Production orders). Everything it does not mention — filters, sort, paging, rows, empty/no-match states, header and navigation — stays exactly as specified in [002_BD](#), which is not edited. Item numbers continue 002_BD's numbering (1–27), so new items start at 28.

System overview

A signed-in Admin or Operator can download the production orders they are looking at on SCR-002 as a CSV file that opens directly in Japanese Excel. The export contains every order that matches the applied filters, in the applied sort order, across all pages (DEC-002), with a fixed set of columns (DEC-003). It reads data only.

Requirement ID	Description	Covered by section
REQ-085	「CSV出力」 downloads all orders matching the applied filters, in the current sort, across all pages	Business flow; §3 item 28; §6 E-30; Actions and business rules
REQ-086	Fixed Japanese columns with values formatted as on screen	§4 M-11–M-15; CSV file layout
REQ-087	Opens correctly in Japanese Excel: UTF-8 with BOM, CRLF, RFC 4180 quoting, timestamped filename	CSV file layout; §4 M-16
REQ-088	Admin/Operator only, same validation as the list, formula-injection neutralised, 10,000-row limit	§5 V-14, V-15; Actions and business rules; Exception flows; Non-functional requirements
REQ-089	Busy, success and error feedback; what will be exported is stated; accessible on PC and SP	§1; §3 items 28–30; §4 M-18; §6 E-30–E-33; Non-functional requirements

Overall configuration and architecture

No new component boundary. The browser calls one new read-only endpoint under the existing `/api/production-orders` path; the .NET backend reuses the list's filter validation and query (002_BD FN-010, FN-011) without paging, writes the rows as CSV and returns them as a file download. The CSV is produced on the server, so filtering, sorting and formatting have one implementation and the browser never has to page through the whole result set. Authentication and roles are WI-001's cookie session and policy, unchanged. The exact endpoint, headers and status codes are fixed in 002_DD-CSV.

Function list

Function ID	Function name	Description	Related requirement ID
FN-041	Export production orders	Return every order matching the given filters in the given sort as one CSV file, refusing the request when more than 10,000 orders match	REQ-085, REQ-088
FN-042	Write CSV safely	Write the header and rows: fixed columns, display formats, UTF-8 with BOM, CRLF, RFC 4180 quoting, and neutralising of text cells a spreadsheet would run as a formula	REQ-086, REQ-087, REQ-088

Function ID	Function name	Description	Related requirement ID
FN-043	Download from the list	Send the screen's applied view to FN-041, save the returned file under its timestamped name, and show progress, success and failure on the screen	REQ-085, REQ-089

Reused unchanged: FN-011 (validate query parameters) and FN-016 (enforce authentication and role) from 002_BD; FN-013 (mark overdue) supplies the 「納期遅れ」 column.

Actors and business flow

Actors: Admin, Operator, signed in (WI-002 DEC-001). Same users as SCR-002; no new permission.

Export the listed orders (UC-022)

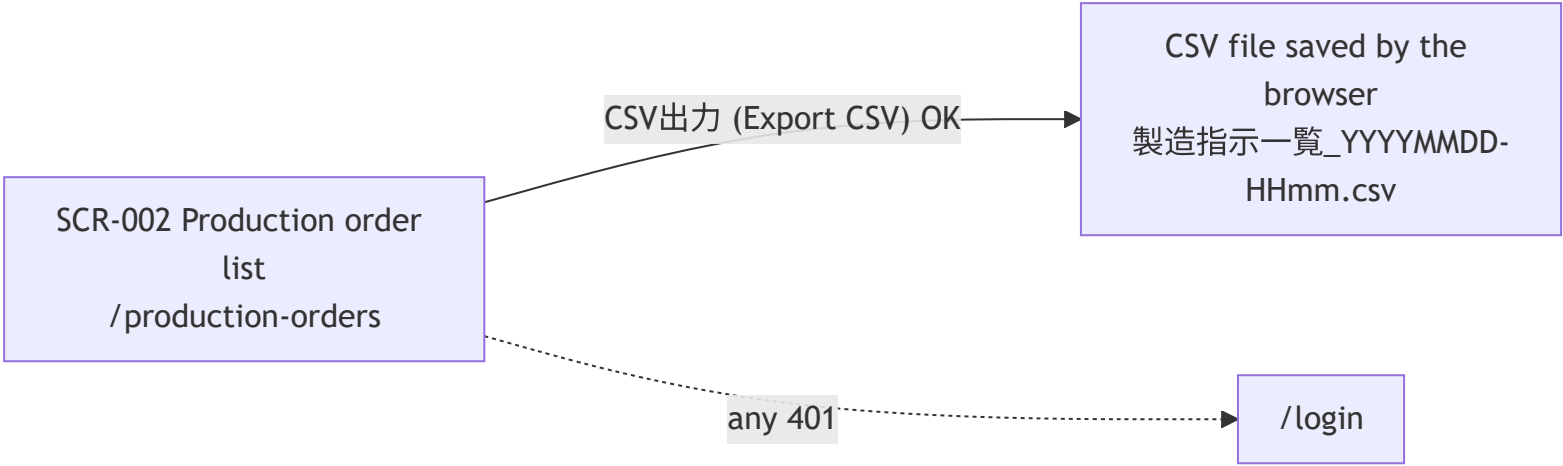
1. User narrows the list with filters and sort as today (002_BD UC-005, UC-006) → the screen shows the result summary, e.g. 「87件中 1～20件」 (1–20 of 87 orders).
2. User presses **CSV出力** (Export CSV) → the system takes the *applied* view from the URL (not unsubmitted edits in the filter panel; the hint under the button, item 30, says 「検索した絞り込み条件の87件を出力します。」 (Exports the 87 orders matching the searched filters.)), ignores page and page size, and requests the export (FN-043). The button shows 「出力中…」 (Exporting…) and is disabled until the request ends.
3. System checks the session, role and parameters (FN-016, FN-011), counts the matches and, if there are at most 10,000, writes them as CSV in the screen's sort order (FN-041, FN-042).
4. Browser saves 製造指示一覧_YYYYMMDD-HHmm.csv (plant time of the export); the screen announces 「CSVファイルを出力しました (87件)。」 (Exported 87 orders to CSV.). The list itself does not change.
5. User opens the file in Excel: Japanese text, dates and quantities display correctly without an import wizard.

Alternative: if more than 10,000 orders match, nothing is downloaded and the screen shows 「出力件数が上限 (10,000件) を超えています。条件を絞り込んでから、もう一度お試しください。」 (The export exceeds the 10,000-order limit. Narrow the filters and try again.). The user narrows the filters and exports again.

Screen list and screen transition

Screen ID	Screen name	Entry point	Exit / next screen
SCR-002	Production Order List (unchanged entries, 002_BD)	/production-orders	「CSV出力」 (Export CSV) → the browser saves a CSV file; the user stays on SCR-002. Session expired / 401 → /login . All other exits as 002_BD

No new screen or route is added. Screen transition for the new action:



Screen design detail

SCR-002 Production Order List — CSV export additions

0-1. Basic information (基本情報)

No	Item	Content	Reference
1	Route / path	/production-orders (unchanged)	002_BD 0-1
2	API base path	/api/production-orders — adds the export endpoint /api/production-orders/export	Exact contract in 002_DD-CSV

No	Item	Content	Reference
3	Character encoding	Screen UTF-8; CSV file UTF-8 with BOM (DEC-004)	CSV file layout
4	Error page / fallback	Export failure: inline message next to the button (item 29); the list stays usable	Exception flows
5	Responsive	Yes — PC ($\geq 640\text{px}$) and SP ($< 640\text{px}$), as 002_BD	§1
6	Authentication required	Yes	REQ-088
7	Authorization / role restriction	Admin , Operator	REQ-088
8	Applicable channel(s)	Single web app — not applicable	

0-2. Page metadata (head)

Not applicable — no change to 002_BD 0-2. The download does not change the page title.

0-3. URL parameters

No new screen parameter. The export sends the screen's applied filter and sort parameters (002_BD 0-3 Nos 1–7: `status` , `productId` , `dueFrom` , `dueTo` , `orderNumber` , `sort` , `dir`) to the export endpoint with the same names and meanings. `page` and `pageSize` are not sent: the export covers all pages (DEC-002).

1. Layout and mockup

Layout-level sketch only; the rendered per-state mockup belongs in 002_DD-CSV. Only the result header area changes; the wireframes draw the rest of the screen in outline for orientation.

1 ProductionManagementAI

製造指示一覧

Admin · ログアウト

ホーム > 製造指示一覧

製造指示一覧

+ 新規製造指示

7-13 絞り込み（ステータス・製品・納期・指示番号） — 002_BD のとおり、変更なし

14 87件中 1～20件

28 CSV出力

15 表示件数 20

30 検索した絞り込み条件の87件を出力します。

29 CSVファイルを出力しました（87件）。

16 指示番号	製品	数量	納期 ▲	ステータス	更新日時
PO-2026-00001	P-1001 — プレーキキャリパー	120 個	2026/09/30	下書き	2026/09/20 10:12
PO-2026-00002	P-1004 — ドライブシャフト	40 個	2026/09/18	進行中	2026/09/19 16:40
PO-2026-00003	P-2002 — 防錆塗料	12.5 kg	2026/10/02	下書き	2026/09/19 11:05

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28 while exporting: 「出力中...」, disabled. 28 is hidden on the empty (24), no-match (25) and loading (26) states.

29 also shows errors (MSG-E024 row limit, MSG-E025 failure, MSG-E020 forbidden) as an alert. Dashed/outlined regions are unchanged (002_BD).

Item No.	Region / element	Notes (behavior, condition)
1–13	Header, breadcrumb, heading, banner, filter panel	Unchanged (002_BD)
14	Result summary	Unchanged; tells the user how many orders the export will contain (「87件中 1～20件」)
15	「表示件数」 (Rows per page)	Unchanged; does not affect the export
28	「CSV出力」 (Export CSV)	New. Secondary (outlined) button with a download icon, in the result header row to the left of item 15. Shown whenever rows are shown (<code>total</code> $\geq$ 1); disabled while the list query or an export is in flight; reads 「出力中…」 (Exporting...) during an export
30	Export hint	New. Small grey text directly under the result header row, always shown with item 28: what the export will contain, from the applied filters and the shown total (M-18). It is also item 28's accessible description. It does not reflect unsubmitted edits in the filter panel; pressing 「検索」 (Search) updates it (DEC-009)
29	Export message	New. One line under item 30: the success announcement (MSG-I009) or an export error (MSG-E024, MSG-E025, MSG-E020). Hidden until an export finishes; cleared by the next export or list query
16–23	Table and paging	Unchanged; the export never changes the page shown

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メニュー

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製造指示一覧

+ 新規製造指示

7-13 絞り込み (2) — 002_BD のとおり

14 87件中 1～20件

15 表示件数 20 ▼

28  CSV出力

30 検索した絞り込み条件の87件を出力します。

29 CSVファイルを出力しました（87件）。

16 並べ替え 納期 ▲

22 PO-2026-00001
P-1001 — ブレーキキャリパー
数量 120 個・納期 2026/09/30
下書き・更新 2026/09/20 10:12

... cards and paging as 002_BD

28: full width, ≥ 44 px; 「出力中...」 while exporting.

Same item numbers as PC. Difference: item 28 takes a full-width row of its own under the summary/ 「表示件数」 row, so the 390 px row does not overflow and the button stays at least 44 px high for touch; items 30 and 29 sit under it, in that order.

2. Content block definition (CMS)

None — no externally managed content blocks.

3. Screen item definition

No	Item (label)	Variable name	Control type	I/O	Data type	Width / length	Initial value	Placeholder	Display condition	Data source	Reference
28	「CSV出力」 (Export CSV)	—	button	I	—	—	Enabled	—	Whenever rows are shown (<code>total</code> ≥ 1); hidden on the empty (24), no-match (25) and loading (26) states; disabled while a list query or export is in flight	Applied view state (URL)	\$6 E-30; \$5 V-14
30	Export hint	—	label	O	text	—	M-18	—	With item 28	Applied view state (URL) + shown <code>total</code>	\$4 M-18; DEC-009
29	Export message	—	label (live region)	O	text	—	Hidden	—	After an export finishes	Export result	\$6 E-31, E-32; MSG-I009, MSG-E020, MSG-E024, MSG-E025

CSV file layout

One header row, then one row per order, in the applied sort order (002_BD V-13 keys; order number ascending as the tie-breaker, as on screen). Columns (DEC-003):

Col	Header (Japanese)	Gloss	Content	Mapping
1	指示番号	Order no.	Order number, e.g. P0-2026-00001	as stored
2	製品コード	Product code	Product SKU	as stored
3	製品名	Product name	Product name	as stored
4	生産ライン	Production line	Assigned line	M-14
5	数量	Quantity	Exact quantity	M-12
6	単位	Unit	Product unit, e.g. 個, kg, m	as stored
7	納期	Due date	Plant-local due date	M-13
8	ステータス	Status	Japanese status label	M-11 (= 002_BD M-05)
9	納期遅れ	Overdue	「納期遅れ」 or empty	M-15 (= 002_BD M-08)

Col	Header (Japanese)	Gloss	Content	Mapping
10	備考	Notes	Order notes, may contain commas, quotes and line breaks	as stored, quoted per FN-042
11	作成日時	Created	Creation timestamp	M-13
12	更新日時	Updated	Last update timestamp	M-13
13	完了日時	Completed	Completion timestamp; empty unless the order was completed	M-13

File format (DEC-004): UTF-8 with a byte-order mark, CRLF line ends, comma separator. A field is enclosed in double quotes when it contains a comma, a double quote, CR or LF, and a double quote inside it is doubled (RFC 4180). A text field that begins with `=`, `+`, `-`, `@`, a tab or a CR gets a leading apostrophe `'` so a spreadsheet shows it as text instead of running it as a formula (FN-042). Numbers and dates are never altered by this rule.

4. Item value mapping

No	Item	Source value	Displayed value	Reference
M-11	Col 8 ステータス	Draft / InProgress / Completed / Cancelled	下書き / 進行中 / 完了 / 取消	002_BD M-05
M-12	Col 5 数量	Stored decimal quantity	Exact decimal with no thousands separator and no trailing zeros, e.g. 120 , 12.5 , 0.125 (kg/m keep up to 3 decimals, WI-006) — so Excel reads it as a number	REQ-086
M-13	Cols 7, 11–13 dates	Stored plant-local date / UTC timestamp	Due date YYYY/MM/DD ; timestamps YYYY/MM/DD HH:mm , 24-hour, in plant time Asia/Tokyo (DEC-008); empty when absent	002_BD M-10; DEC-008
M-14	Col 4 生産ライン	Assigned line or none	{code} – {name} , with 「(使用停止)」 (retired) appended for a retired line, as on screen; empty when no line is assigned	WI-009
M-15	Col 9 納期遅れ	Due date, status, plant today at export time	「納期遅れ」 when due date < plant today and status is Draft or InProgress ; otherwise empty	002_BD M-08, FN-013
M-16	Download file name	Plant-local time of the export	製造指示一覧_YYYYMMDD-HHmm.csv , e.g. 製造指示一覧_20261007-1345.csv ; an ASCII fallback production-orders_YYYYMMDD-HHmm.csv for clients that cannot take a Japanese name	REQ-087
M-18	Item 30 export hint	Applied filters (URL) and shown total	With any filter applied: 「検索した絞り込み条件の{total}件を出力します。」 (Exports the {total} orders matching the searched filters.); with no filter: 「すべての製造指示{total}件を出力します。」 (Exports all {total} production orders.). Sort is not mentioned; rows follow the list's order	DEC-009
M-17	Item 29 success	Number of exported rows	「CSVファイルを出力しました ({count} 件) 。」 (Exported {count} orders to CSV.)	MSG-I009

5. Validation rules

The export endpoint repeats every list check (002_BD V-09–V-13 for the filter and sort parameters) and is authoritative. The screen only exports a view that the list has already loaded successfully, so in normal use these checks pass; a crafted request that fails them gets the same 400 as the list.

No	Item	Check content	Validation rule	Check condition	Error message (ID)	Reference
V-14	(28) Export	Row limit	At most 10,000 orders match the filters (DEC-005). The screen checks the shown <code>total</code> first and shows the message without a request; the server counts again and refuses if over	On click (client); on export (server)	MSG-E024 「出力件数が上限（10,000件）を超えています。条件を絞り込んでから、もう一度お試しください。」 (The export exceeds the 10,000-order limit. Narrow the filters and try again.)	REQ-088
V-15	Export parameters	Same rules as the list	002_BD V-09–V-13 for <code>status</code> , <code>productId</code> , <code>dueFrom</code> , <code>dueTo</code> , <code>orderNumber</code> , <code>sort</code> , <code>dir</code> ; <code>page</code> and <code>pageSize</code> are not accepted parameters of the export	On export (server)	As 002_BD: MSG-E002, MSG-E015–MSG-E019	REQ-088

Messages. New IDs continue the single catalog (`messages.ts`), which ends at MSG-E023 and MSG-I008:

ID	Text (draft)	Shown when	New or reused
MSG-I009	「CSVファイルを出力しました（{count}件）。」 (Exported {count} orders to CSV.)	Export succeeded (item 29)	new
MSG-E024	「出力件数が上限（10,000件）を超えています。条件を絞り込んでから、もう一度お試しください。」 (The export exceeds the 10,000-order limit. Narrow the filters and try again.)	V-14 fails	new
MSG-E025	「CSV出力に失敗しました。もう一度お試しください。」 (The CSV export failed. Try again.)	Network, server or unexpected error during the export	new — more specific than MSG-E013, so the user knows the list itself is fine
MSG-E020	「製造指示を閲覧する権限がありません。」 (You don't have permission to view production orders.)	403 on the export	reused from 002_BD
MSG-E002, MSG-E015–MSG-E019	as 002_BD	V-15 fails (crafted request)	reused from 002_BD

6. Item events

No	Item	Event	Event content	Reference
E-30	(28) 「CSV出力」 (Export CSV)	click / Enter / Space	If the shown <code>total</code> > 10,000, show MSG-E024 in item 29 and stop (V-14). Otherwise clear item 29, set the button to 「出力中…」 (Exporting…) and disabled, and request the export with the applied filter and sort parameters (0-3). Focus stays on the button	FN-043; REQ-085

No	Item	Event	Event content	Reference
E-31	Export	success	The browser saves the file under the name in M-16; item 29 announces MSG-I009 with the exported row count; the button returns to 「CSV出力」. The list, URL and page are unchanged	FN-043; REQ-089
E-32	Export	failure	401 → the existing session handling sends the user to <code>/login</code> . 403 → MSG-E020; row limit → MSG-E024; 400 → the list's messages for the rejected parameters; anything else → MSG-E025. Shown in item 29 as an alert; nothing is saved; the button is enabled again	REQ-088, REQ-089
E-33	List query	starts / finishes (Search, Clear, sort, page, page size, Back/Forward)	Items 28 and 30 are hidden while the query runs (002_BD item 26) and item 29 is cleared, so an old message never describes a new view; when the query finishes, item 30 is rebuilt from the new applied filters and total	REQ-089

No event writes order data.

7. External identity linkage

None — no external identity linkage.

Actions and business rules

Action	Trigger	Business rule	Related requirement ID
Call the export API	Item 28	Requires a valid session (else 401) and role <code>Admin</code> or <code>Operator</code> (else 403), enforced server-side as for the list	REQ-088
Select rows	Export request	Same filters, combined with AND, as the list query with the same parameters; absent filters place no restriction; no paging	REQ-085
Order rows	Export request	The requested <code>sort / dir</code> (default <code>dueDate</code> ascending), order number ascending as the tie-breaker — identical to the screen	REQ-085
Limit size	Export request	Refuse with MSG-E024 when more than 10,000 orders match; never send a partial file	REQ-088; DEC-005
Write cells	Each row	Display formats M-11–M-15; formula-like text neutralised; quoting per RFC 4180	REQ-086–REQ-088
Read only	Any export	Changes no order data and does not alter the screen's view state	REQ-085

Success and exception flows

Flow	Trigger condition	System behavior	Resulting state
Success — filtered export	Rows shown, ≤ 10,000 match	File with header + all matching rows, in screen order	File saved; MSG-I009; list unchanged
Success — unfiltered export	No filter set, ≤ 10,000 orders exist	Every order exported	As above

Flow	Trigger condition	System behavior	Resulting state
Not offered — nothing to export	Empty state (24) or no-match state (25)	Item 28 is not shown	No action possible
Exception — over the limit	> 10,000 match (shown total, or server count)	Request not sent, or refused by the server; no file	MSG-E024 in item 29; list unchanged
Exception — unauthenticated	Session expired	401, no data	Redirect to <code>/login</code>
Exception — forbidden	Signed in without <code>Admin / Operator</code> (crafted request)	403, no data	MSG-E020
Exception — invalid parameter	V-15 fails (crafted request)	400, no data	The list's validation messages in item 29
Exception — server/network error	Unexpected failure, or the connection drops	No file is saved; a partial download is discarded	MSG-E025 in item 29; button enabled again
Data changed during export	Another user edits an order between the list query and the export	The export reflects the data at export time; its count may differ from the shown total	MSG-I009 states the exported count

Data design overview

No new entity, column, index or migration, so no 002_DB amendment is needed. The export reads **ProductionOrder** (order number, quantity, due date, status, notes, created/updated/completed timestamps) joined to its **Product** (SKU, name, unit) and, when assigned, its **ProductionLine** (code, name, active state). The filter and sort run on the indexes of [002_DB](#) and WI-004; the export only drops `OFFSET/LIMIT` and adds a count check capped by DEC-005. Notes are included in the export although the list does not show them.

External interfaces

None — only this application's own backend API. The file is consumed by spreadsheet tools such as Microsoft Excel, which is why the file format above is fixed.

Non-functional requirements

- **Security:** server-side role check as for the list (REQ-088). Parameters are allow-listed and bound exactly as the list's (002_BD non-functional requirements), so no client value reaches SQL. Formula injection (CWE-1236) is neutralised in every text cell. The response is marked not cacheable (`no-store`) so a shared

- machine's browser cache does not keep the file. The export contains no PII or secrets, but it does contain all matching orders' notes; access is the same as reading each order on SCR-001. CSRF does not apply to this read-only GET. Reviewed against ai/checklists/security-review.md in 002_DD-CSV.
- **Accessibility (WCAG 2.2 AA):** item 28 is a real button with a visible text label (the icon is decorative); its accessible description is the visible hint, item 30 (M-18); its busy state is exposed (aria-busy /disabled with the 「出力中…」 label); item 29 is a polite live region for success and an alert for errors; focus stays on the button; the SP button is at least 44 px high; text never relies on colour.
 - **Performance:** the server streams rows instead of building the whole file in memory. Target: 10,000 rows within 5 seconds on the local Compose stack; the 124-order demo data exports in well under a second. Measured in the implementation revision.
 - **Observability:** each export logs one structured entry (user, filter and sort parameters, row count, duration, outcome) and is traced like the list query; order contents are not logged. Exact names are fixed in 002_DD-CSV.
 - **Availability:** inherits project defaults.

Open questions and linked DD

Question	Linked DD section	Status
Scope, columns, encoding, row limit	—	answered in decisions.md (DEC-002–DEC-005)
Timestamps in plant time vs. browser time	—	answered in decisions.md (DEC-008) — plant time
HTTP status and Problem Details code for the row limit	002_DD-CSV API	open — fixed in 002_DD-CSV
How the browser receives the row count for MSG-I009 (response header)	002_DD-CSV API	open — fixed in 002_DD-CSV
Streaming approach and how a mid-stream failure is detected by the browser	002_DD-CSV processing	open — fixed in 002_DD-CSV